# Information Retrieval

## Retrieval Models II: BM25, LM and evaluation

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# Roadmap of this lecture

- A recap
- BM25
- Language model

- IR system evaluation

# Best Match Okapi Model (BM25)

## Building on the probabilistic model: Okapi weighting

- Okapi system is based on the probabilistic model
- BIRM does not perform as well as the vector space model
  - does not use term frequency (*tf*) and document length (*dl*)
  - hurt performance on long documents
- What Okapi does:
  - add a *tf* component like in the vector space model
  - separate document and query length normalization
  - several tuning constants, which depend on the collection

# Best Match Okapi Model (BM25)

## BM25 (Best-match Okapi weight)

$$R(d,q) = BM25(d,q) =$$

$$\sum_{t \in q} \left( w_t \cdot \frac{(k_1+1)tf(t,d)}{K + tf(t,d)} \cdot \frac{(k_3+1)tf(t,q)}{k_3 + tf(t,q)} \right) + k_2 \cdot |q| \cdot \frac{avgdl - dl}{avgdl + dl}$$

$$K = k_1((1-b) + (b \cdot dl)/avdl)$$

| | |
|---|---|
| $w_t$: | term weight based on relevance feedback |
| $tf(t, d)$, $tf(t, q)$: | within term frequencies - document and query |
| $k_1$, $k_2$, $k_3$, $b$: | tuning parameters |
| $dl$, $avgdl$: | document length and average document length |

# Best Match Okapi Model (BM25)

## BM25 - Parameters

$$\sum_{t \in q} \left( w_t \cdot \frac{(k_1+1)tf(t,d)}{K + tf(t,d)} \cdot \frac{(k_3+1)tf(t,q)}{k_3 + tf(t,q)} \right) + k_2 \cdot |q| \cdot \frac{avgdl - dl}{avgdl + dl}$$

$$K = k_1((1-b) + (b \cdot dl) / avdl)$$

$k_1$ : governs the importance of within document frequency $tf(t, d)$

$k_2$: compensation factor for the high within document frequency values in large documents

$k_3$: governs the importance of within query frequency $tf(t, q)$

$b$: relative importance of within document frequency and document length

The theoretical basis for the Okapi formula is the use of Poisson distributions to model within document frequency in relevant documents, and in non-relevant documents (not discussed here).

# Best Match Okapi Model (BM25)

- A common simplified form
  - $k_2$ =0; $k_3$ large; $b$ closer to 1

$$RSV^{BM25} = \sum_{i \in q} \log \frac{N}{df_i} \cdot \frac{(k_1+1)tf_i}{k_1((1-b)+b\frac{dl}{avdl})+tf_i}$$

- Typically, $k_1$ is set around [1.2,2] and $b$ around 0.75

# Best Match Okapi Model (BM25)

- Many formulations and interpretations of BM25
- $\log \frac{N}{df_i}$ : the IDF term
- Can be re-written as $\log \frac{N - n(q_i) + 0.5}{n(q_i) + 0.5}$
- $N$ is the total number of documents in the collection
- $n(q_i)$ is the number of documents containing $q_i$

# Best Match Okapi Model (BM25)

## BM25 (Best-match Okapi weight)

- Experiments show:
  - For instance ( with $k_1$ = 1 and $b$ = 0.75):

$$BM25(d,q) = \sum_{t \in q} (w_t \cdot \frac{tf(t,d)}{K + tf(t,d)})$$

  - $K = 0.25 + (0.75\ dl)/avdl$
- In experiments, Okapi weights give the best performance. BM25 often used as baseline model in retrieval experiments.

# Best Match Okapi Model (BM25)

## Summary

- The vector space model (VSM) is a “geometrical” model; it is viewed as “basic”. Established in the 70s.
- The BIRM is one of the important pieces of IR theory.
  - Ranking based on the probability of relevance is optimal with respect to a cost function where the costs for reading relevant documents are low and the costs for reading non-relevant documents are high; probability ranking principle [Robertson 1977].
- BM25 Okapi model is often the most “effective” model, the model to “beat” in retrieval experiments.
- BM25F (BM25 Field) – take document structure and anchor text into account

## Generative Probabilistic Models

- The **generative approach**

  A generator which produces events/tokens with some probability

  - URN Metaphor: a bucket of different colour balls (10 red, 5 blue, 3 yellow, 2 white)
    - What is the probability of drawing a yellow ball?
    - What is the probability of drawing (with replacement) a red ball and a white ball?
  - IR Metaphor: Documents are urns, full of tokens (balls) of (in) different terms (colours)

## Generative Probabilistic Models

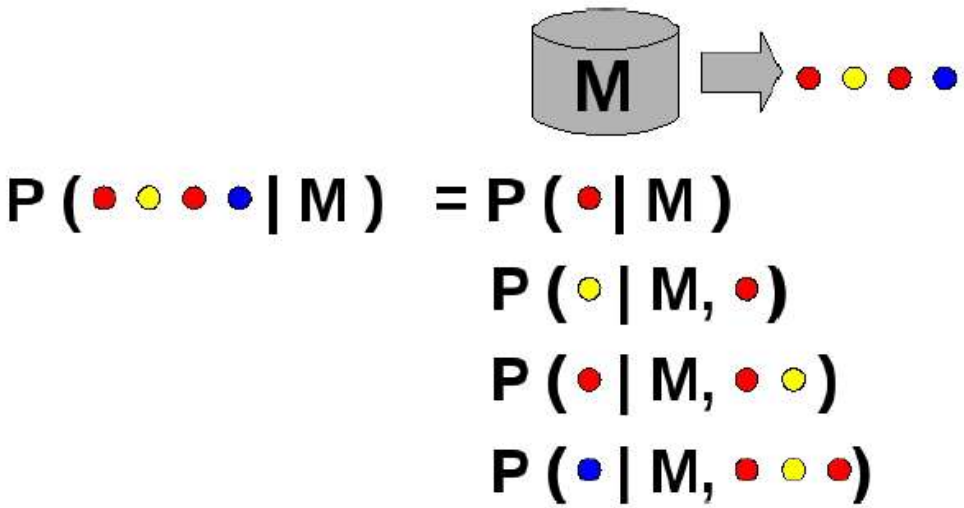
- What is the probability of producing the query from a document? $P(q|d)$
  - Referred to as the query-likelihood
- The query is generated as a representative of the “ideal” document
  - System’s task is to estimate for each of the documents in the collection, which is most likely to be the “ideal” document

# Generative Probabilistic Models

## Generative Models - Language model

A statistical model for generating data

- Probability distribution over samples for a given language (document)
- M –> $t_1\ t_2\ t_3\ t_4$



## Generative Probabilistic Models

- Assumptions:
  - The probability of a document being relevant is strongly correlated with the probability of a query given a document, i.e. $P(q|r)$ is correlated with $P(q|d)$
  - User has a reasonable idea of the terms that are likely to appear in the “ideal” document
  - Users query terms can distinguish the ideal document from the rest of the corpus

# Language Models in IR

## Statistical Language Models

- (Statistical) language models (LM) have been widely used for speech recognition and language (machine) translation for more than thirty years
- However, their use for information retrieval started only in 1998 [Ponte and Croft, SIGIR 1998]
  - Basically, a query is considered generated from an “ideal” document that satisfies the information need
  - The system’s job is then to estimate the likelihood of each document in the collection being the ideal document and rank them accordingly (in decreasing order)

# Language Models in IR

## Statistical Language Models

- What is LM Used for ?
  - Speech recognition
  - Spelling correction
  - Handwriting recognition
  - Optical character recognition
  - Machine translation
  - Document classification and routing
  - Information retrieval …

## Language Models in IR

- Let us assume we point blindly, one at a time, at 3 words in a document
- What is the probability that I, by accident, pointed at the words “Master”, “computer”, and “Science”?
- Compute the probability, and use it to rank the documents.

# Language Models in IR

## Statistical Language Models

- A probabilistic mechanism for "generating" a piece of text
  - Define a distribution over all possible word sequences

$$T = t_1 t_2 \ldots t_L$$

$$P(T) = ?$$

  - Used LM to quantify the accept ability of a given word sequence

# Language Models in IR

## Query-Likelihood Language Models

- Criterion: Documents are ranked based on Bayes (decision) rule

$$P(D \mid Q) = \frac{P(Q \mid D) \cdot P(D)}{P(Q)}$$

  - $P(Q)$ is the same for all documents, and can be ignored
  - $P(D)$ might have to do with authority, length, genre, etc.
    - There is no general way to estimate it
    - Can be treated as uniform across all documents

# Language Models in IR

## Query-Likelihood Language Models

- Documents can therefore be ranked based on

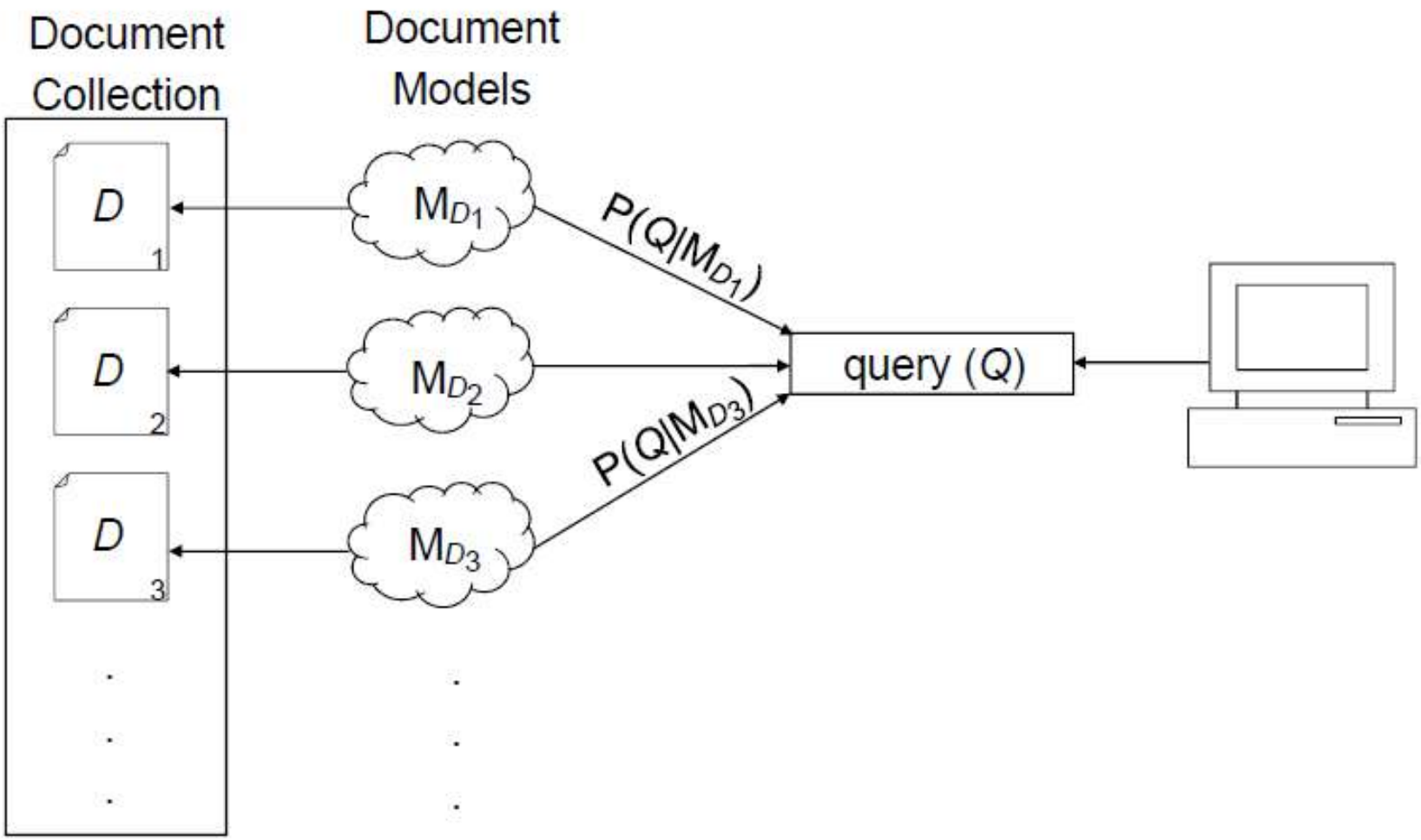
$$P(Q \mid D) \text{ or denoted as } P(Q \mid M_D)$$

Document models

  - The user has a prototype (ideal) document in mind, and generates a query based on words that appear in this document
  - A document $D$ is treated as a model $M_D$ to predict (generate) the query

# Language Models in IR

## Schematic Depiction for Query-Likelihood Approach



# Language Models in IR

## Types of language models – urn metaphor

$$P(\bullet\,\circ\,\bullet\,\bullet)$$

$$= P(\bullet)\,P(\circ \mid \bullet)\,P(\bullet \mid \bullet\,\circ)\,P(\bullet \mid \bullet\,\circ\,\bullet)$$

- Unigram Models

$$P(\bullet)\,P(\circ)\,P(\bullet)\,P(\bullet)$$

- Bigram Models

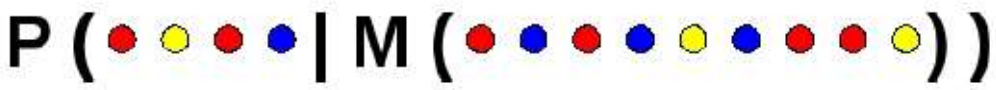
$$P(\bullet)\,P(\circ \mid \bullet)\,P(\bullet \mid \circ)\,P(\bullet \mid \bullet)$$

- There are others . . .
  - Most language-modelling work in IR has used unigram models
  - IR does not directly depend on the structure of sentences

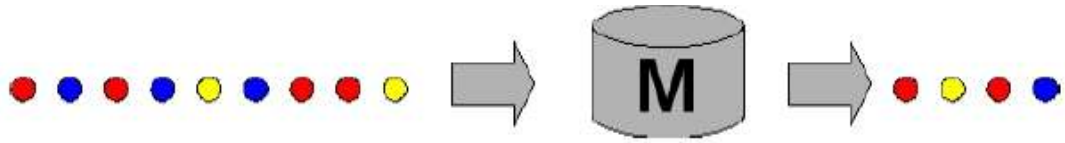
# Language Models in IR

## The fundamental problem

- Usually, we do not know the model M, but have a sample representative of that model



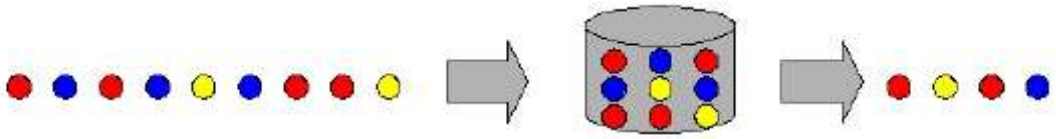

- First estimate a model from a sample
- Then compute the observation probability



# Language Models in IR

## Unigram Model - Example

(Urn metaphor)

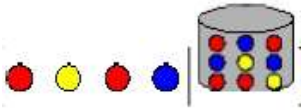
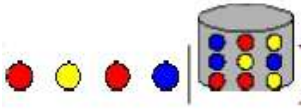
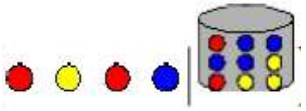
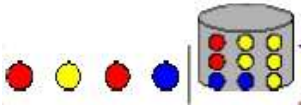


- P(● ○ ● ●) ~ P (●) P (○) P (●) P (●)

  = 4/9 * 2/9 * 4/9 * 3/9

# Language Models in IR

## Unigram Model - Example:
## Ranking documents with unigram models

- Rank models (documents) by probability of generating the query
- Q:
- $P($  $) =$
- $P($  $) =$
- $P($  $) =$
- $P($  $) =$

# Language Models in IR

## Build Document Models: *n*-grams

- Multiplication (Chain) rule

$$P(t_1 t_2 \ldots . t_L) = P(t_1) P(t_2 \mid t_1) P(t_3 \mid t_1 t_2) \cdots P(t_L \mid t_1 t_2 \ldots . t_{L-1})$$

- *n*-gram assumption
  - Unigram Models (Assume word independence)

$$P(t_1 t_2 \ldots . t_L) = P(t_1) P(t_2) P(t_3) \cdots P(t_L)$$

    - Each word occurs independently of the other words
    - The so-called “bag-of-words” model (e.g., how to distinguish “street market” from “market street)
  - Bigram Models

$$P(t_1 t_2 \ldots . t_L) = P(t_1) P(t_2 \mid t_1) P(t_3 \mid t_2) \cdots P(t_L \mid t_{L-1})$$

# Language Models in IR

## Unigram Model - Standard LM Approach

- Assume that query terms are drawn identically and independently from a document (unigram models)

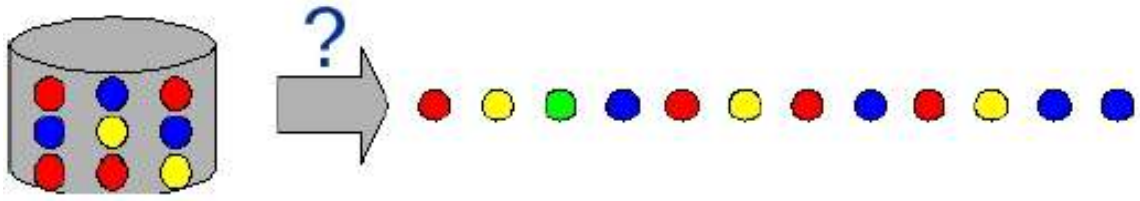
$$P(q \mid d) = \prod_{t \in q} P(t \mid d)^{n(t,q)}$$

(where $n(t, q)$ is the number of term $t$ in query $q$)

- Maximum Likelihood Estimate of $P(t|d)$
  - Simply use the number of times the query term occurs in the document divided by the total number of term occurrences.
- Problem: **Zero Probability (frequency) Problem**

## Unigram Model - The Zero-probability Problem

- Suppose some event not in sample (document)
  - Model will assign zero probability to that event
  - And to any set of events involving the unseen event
- Happens frequently with language
- It is incorrect to infer zero probabilities
  - Especially when dealing with incomplete samples



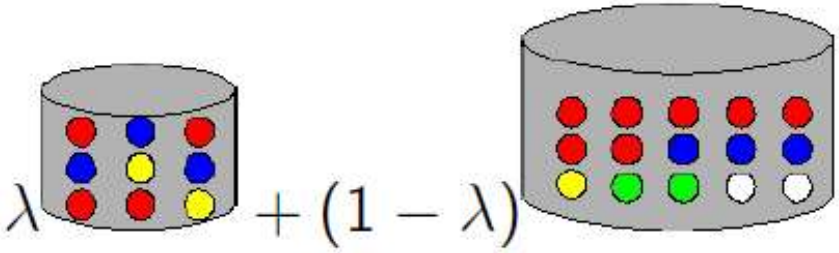
# Language Models in IR

## Unigram Model - Smoothing

- Standard approach is to use the probability of a term $P(t)$ to smooth the document model, thus

$$P(t \mid q_d) = \lambda P(t \mid d) + (1-\lambda) P(t)$$

- Urn metaphor:



# Language Models in IR

## Unigram Model - Document Models

- Idea: shift part of probability mass to unseen events
- Interpolation with background (General English in our case)
  - Reflects expected frequency of events
  - Plays role of IDF in LM

$$P(t \mid q_d) = \lambda P(t \mid d) + \boxed{(1-\lambda)P(t)}$$

Background

## Unigram Model - Estimating Document Models

- Basic Components
  - Probability of a term given a document (maximum likelihood estimate)

$$P(t \mid d) = \frac{n(t,d)}{\sum_{t'} n(t',d)}$$

Foreground model

  - Probability of a term given the collection

$$P(t) = \frac{\sum_{d} n(t,d)}{\sum_{t'} \sum_{d'} n(t',d')}$$

Background model

  - $n(t, d)$ is the number of times term $t$ occurs in document $d$

## Unigram Model - Estimating Document Models

- Example of Smoothing methods
  - Laplace

$$P(t \mid q_d) = \frac{n(t,d) + \alpha}{\sum_{t'} n(t',d) + \alpha \mid T \mid}$$

$|T|$ is the number of term in the vocabulary

  - Jelinek-Mercer

$$\lambda \cdot P(t \mid d) + (1-\lambda) \cdot P(t)$$

  - Dirichlet

$$\frac{|d|}{|d| + \mu} \cdot P(t \mid d) + \frac{\mu}{|d| + \mu} \cdot P(t)$$

# Language Models in IR

## Unigram Model
## Language Models - Implementation

We assume the following LM (**Jelinek-Mercer** smoothing):

$$P(q = t_1, t_2, \ldots, t_n \mid d) = \prod_{i=1}^{n} ((1-\lambda) \cdot P(t_i) + \lambda \cdot P(t_i \mid d))$$

It can be shown that the above leads to:

$$P(q = t_1, t_2, \ldots, t_n \mid d) \propto \sum_{i=1}^{n} \log(1 + \frac{\lambda \cdot P(t_i \mid d)}{(1-\lambda) \cdot P(t_i)})$$

for ranking purpose (again use log to obtain summation)

# Language Models in IR

## Unigram Model: example

$$P(q = t_1, t_2, \ldots, t_n \mid d) = \prod_{i=1}^{n} ((1-\lambda) \cdot P(t_i) + \lambda \cdot P(t_i \mid d))$$

Suppose the document collection contains two documents:

- *d*1: *Xyzzy reports a profit but revenue is down*
- *d*2: *Quorus narrows quarter loss but revenue decreases further*

The model will be unigram models from the documents and collection, mixed with $\lambda = 1/2$.

Suppose the query is *revenue down*. Then:

- $P(q|d1) = ?$
- $P(q|d2) = ?$

ranking *d*1 and *d*2

# Language Models in IR

## Unigram Model - Implementation as vector product

Redefine:

$$P(t) = \frac{df(t)}{\sum_{t'} df(t')} \qquad P(t|d) = \frac{tf(t,d)}{\sum_{t'} tf(t',d)}$$

$$\text{score}(t,d) = \log(1 + \frac{tf(t,d) \cdot \sum_{t'} df(t')}{df(t) \cdot \sum_{t'} tf(t',d)} \cdot \frac{\lambda}{1-\lambda})$$

$$\text{score}(q,d) = \sum_{t \in q} score(t,d)$$

# Language Models in IR

## Unigram Model - Document Priors

- Remember $P(d|q) = P(q|d)P(d)/P(q)$
- $P(q)$ is typically assumed to be uniform so is usually ignored
- $P(d)$ provides an interesting avenue for encoding a priori knowledge about the document
  - Document length (longer doc → more relevant)
  - Average Word Length (bigger words → more relevant)
  - Time of publication (newer doc → more relevant)
  - Number of web links (more in links → more relevant)
  - PageRank (more popular → more relevant)

# Language Models in IR

## Unigram Model - “Language Modelling”

- Not just “English”
- But also, the language of
  - author
  - newspaper
  - text document
  - image
  - structure
  - …..

# Language Models in IR

## Summary LM

- Approach based on “probability” of relevance (like BIRM) but RSV is based on $P(q|d)$ and not $P(d|q, r)$
- Based on the probability that a term occurs in a sequence of terms.
- BIRM is based on the probability that term does or does not occur in a set of (retrieved) documents
- Relation to IDF:
  - Applying the logarithm after dividing by $P(t)$, $t$ in $q$

$$\log\left(\frac{(1-\lambda)+\lambda\cdot P(t|d)}{P(t)}\right) \propto \mathrm{IDF}(t)$$

# Language Models in IR

## Summary LM

- A query *q* is viewed as a translation or distillation from a Document *d*
  - That is, the similarity measure is computed by estimating the probability that the query would have been generated as a translation of that document
  - Assumption of context-independence (the ability to handle the ambiguity of word senses is limited)

# Language Models in IR

## Summary - References


- Ponte and Croft, A language modelling approach to information retrieval, ACM-SIGIR 1998
- Hiemstra and de Vries, Relating the new language models of information retrieval to the traditional retrieval models, CTIT Technical Report, 2000
- Liu and Croft, Statistical Language Modelling for Information Retrieval, Annual Review of Information Science and Technology, 2003
- Zhai and Lafferty, A study of smoothing methods for language models applied to ad hoc information retrieval, ACM TOIS, 2004
- Larenko and Croft, Relevance Based Language Models, ACM SIGIR 2001
- Croft and Lafferty (eds), Language Modelling for Information Retrieval 2004

# Evaluating an IR system

## System-oriented evaluation

- Test collection
  - Benchmark (data set) upon which effectiveness is measured and compared
  - Data that tell us for a given query what are the relevant documents.
- Measuring effectiveness has been the most predominant in IR evaluation
  - **recall** of the system: proportion of **relevant** documents retrieved
  - **precision** of the system: proportion of the retrieved documents that are actually **relevant**
- Looking at these two aspects is part of what is called **system-oriented evaluation**

# Test Collections

## Components and why do we need them?

- Typical components:
  - **Document collection**, i.e. the document themselves. This depends on the task, e.g. evaluating web retrieval requires a collection of HTML documents
  - **Queries / requests**, which simulate real-user information needs
  - **Relevance judgements**, i.e. stating for a query the relevant documents
- To compare the performance of different retrieval techniques
  - Usually use multiple measures to get different views of performance
  - Usually test with multiple collections as performance is collection dependent

# Test Collections

## Relevance judgement ‘groundtruth’ creation

- Manual method:
  - Every document in the collection is judged against every query by one of several judges (human assessors)
  - This is feasible for small document collection.
- Pooling method (used for large document collection):
  - The queries are run against several IR systems first
  - The top, for example 100, documents retrieved by each system are pooled together.
  - The pool is then judged for relevance (by human assessors)

# Test Collections

## TREC (Text REtrieval Conference)

- Text REtrieval Conference
- Established in 1992 to evaluate large-scale IR
  - Retrieving documents from a gigabyte collection
- Has run continuously since then
- Run by NIST (National Institute of Standards and Technology) Information Access Division
- Started with 25 participating organisations; now many groups from many countries
- Proceedings online http://trec.nist.gov

# Test Collections

## Example (excerpt) of a TREC document

```
<doc>
<docno> WSJ880406-0090 </docno>
<hl> AT&T Unveils Services to Upgrade Phone Networks
Under Global Plan </hl>
<author> Janet Guyon (WSJ Sta) </author>
<dateline> New York </dateline>
<text>
American Telephone & Telegraph Co. introduced the
   rest of a new generation of phone services with
   broad ...
</text>
</doc>
```

# Test Collections

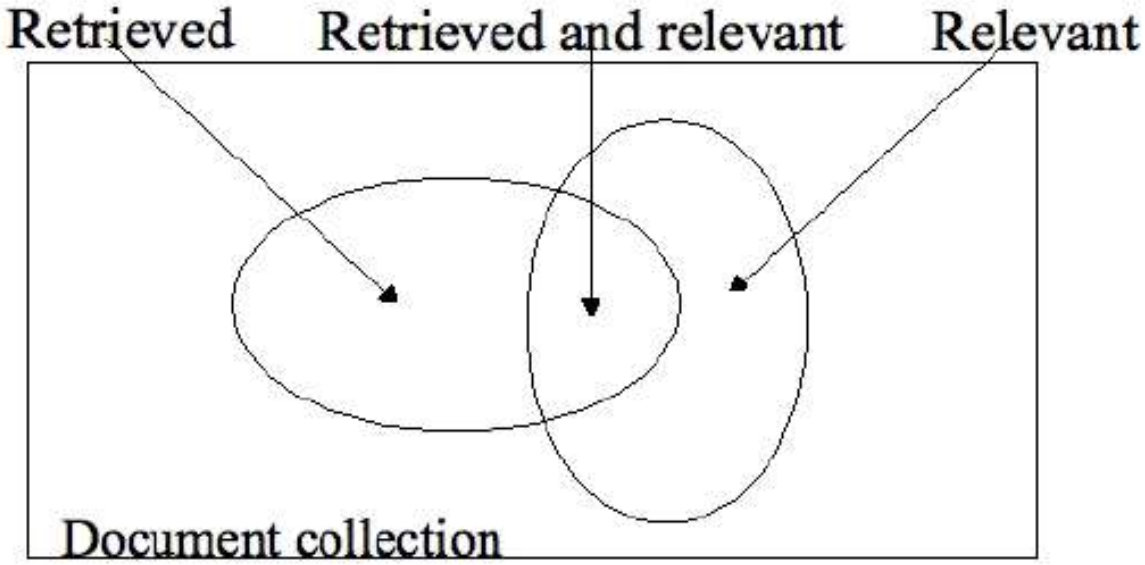
## Example (excerpt) of a TREC topic

```
<top>
<num> Number: 168 </docno>
<title> Topic: Financing AMTRAK
<desc> Description
A document will address the role of the Federal Government
   in financing the operation of the National Railroad
   Transportation Corporation (AMTRAK)
<nar> Narrative:
A relevant document must provide information on the
   government's responsibility to make AMTRAK an
   economically viable entity. It could also discuss the
   privatisation of AMTRAK. Documents comparing government
   subsidies given to air and bus transportation with those
   provided to AMTRAK would also be relevant.
</top>
```

# Precision and Recall

## Recall / Precision

For a given query, the document collection can be divided into three sets: the set of retrieved documents, the set of relevant documents, and the rest of the documents.




Note: knowing which documents are relevant comes from the test collection

# Precision and Recall

## Recall / Precision

In the ideal case, the set of retrieved documents is equal to the set of relevant documents. However, in most cases, the two sets will be different. This difference is formally measured with precision and recall.

$$Precision = \frac{\text{number of relevant documents retrieved}}{\text{number of documents retrieved}}$$

$$Recall = \frac{\text{number of relevant documents retrieved}}{\text{number of relevant documents}}$$

# Precision and Recall

## Recall / Precision

We can say that Precision corresponds to **correctness** and Recall to **completeness**.

Users in different scenarios may have different preference between the two measures. For example:

- Precision-oriented searches dominate e.g. web search. In general, in web searches users want some relevant document high in the ranked list of results (but this depends on the search task, etc.)
- Recall-oriented searches are important in domains where completeness of relevant documents is important, e.g. in the legal domain, in cases of considering patents for filing, etc. - in cases where it is bad to overlook some relevant items.

# Precision and Recall

## Recall / Precision

- Let us assume that for a given query, the following documents are relevant (10 relevant documents)

  {d3, d5, d9, d25, d39, d44, d56, d71, d89, d123}

- Now suppose that the following documents are retrieved for that query:

| rank | doc | precision | recall | rank | doc | precision | recall |
|---|---|---|---|---|---|---|---|
| 1 | d123 | 1/1 | 1/10 | 8 | d129 | | |
| 2 | d84 | | | 9 | d187 | | |
| 3 | d56 | 2/3 | 2/10 | 10 | d25 | 4/10 | 4/10 |
| 4 | d6 | | | 11 | d48 | | |
| 5 | d8 | | | 12 | d250 | | |
| 6 | d9 | 3/6 | 3/10 | 13 | d113 | | |
| 7 | d511 | | | 14 | d3 | 5/14 | 5/10 |

# Precision and Recall

## Recall / Precision

- For each query, we obtain pairs of recall and precision values
  - In our example, we would obtain (1/10, 1/1) (2/10, 2/3) (3/10, 3/6) (4/10, 4/10) (5/10, 5/14) . . . which are usually expressed in % (10%, 100%) (20%, 66.66%) (30%, 50%) (40%, 40%) (50%, 35.71%) . . .
  - This can be read for instance: at 20% recall, we have 66.66% precision; at 50% recall, we have 35.71% precision

| rank | doc | precision | recall | rank | doc | precision | recall |
|---|---|---|---|---|---|---|---|
| 1 | d123 | 1/1 | 1/10 | 8 | d129 | | |
| 2 | d84 | | | 9 | d187 | | |
| 3 | d56 | 2/3 | 2/10 | 10 | d25 | 4/10 | 4/10 |
| 4 | d6 | | | 11 | d48 | | |
| 5 | d8 | | | 12 | d250 | | |
| 6 | d9 | 3/6 | 3/10 | 13 | d113 | | |
| 7 | d511 | | | 14 | d3 | 5/14 | 5/10 |

# Precision and Recall

## Averaging Recall/Precision values

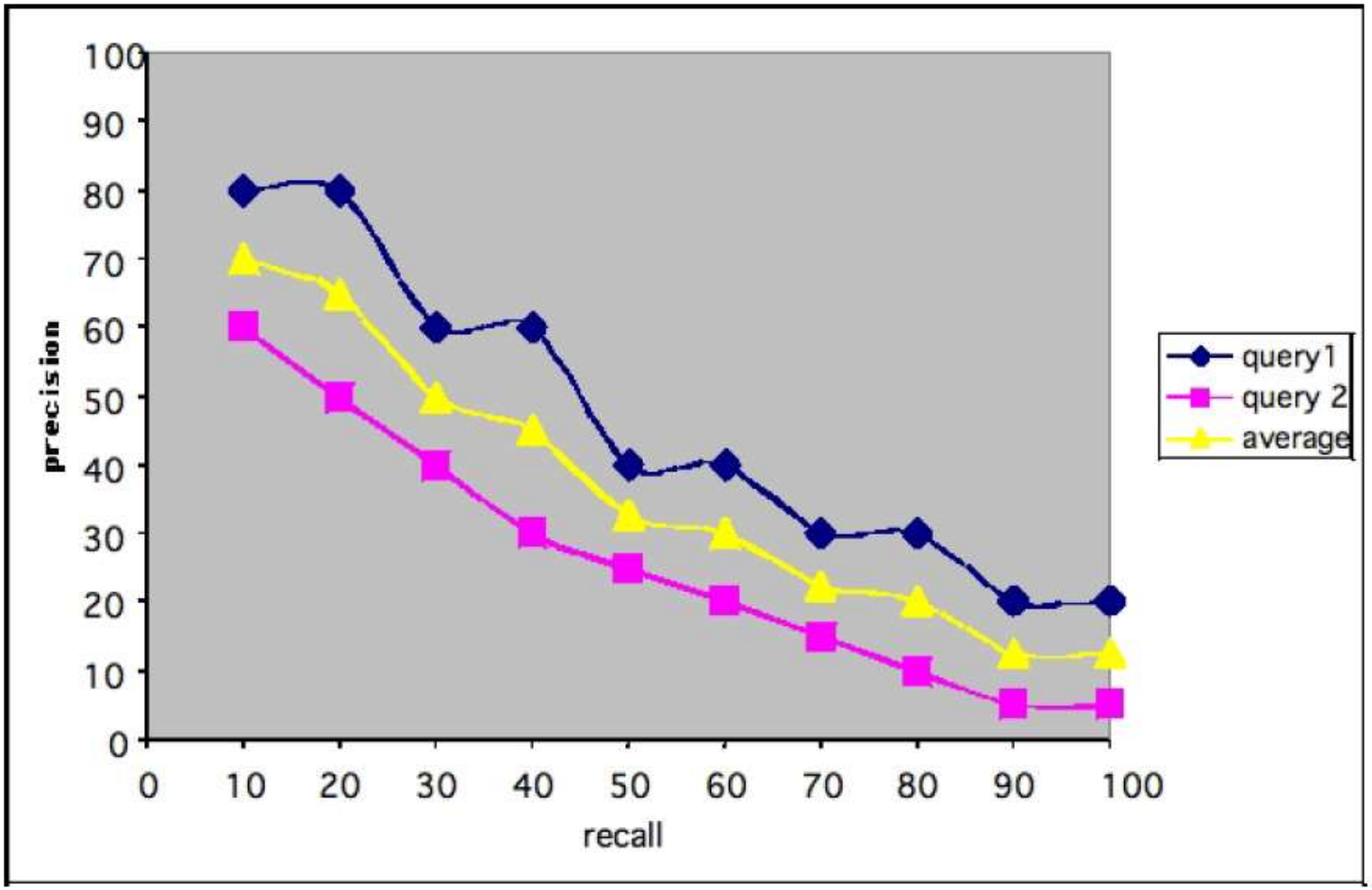
- Hard to compare precision and recall graphs or tables for individual queries (too much data)

| Recall in % | Precision in % | | |
|---|---|---|---|
| | Query 1 | Query 2 | Average |
| 10 | 80 | 60 | 70 |
| 20 | 80 | 50 | 65 |
| 30 | 60 | 40 | 50 |
| 40 | 60 | 30 | 45 |
| 50 | 40 | 25 | 32.5 |
| 60 | 40 | 20 | 30 |
| 70 | 30 | 15 | 22.5 |
| 80 | 30 | 10 | 20 |
| 90 | 20 | 5 | 11.5 |
| 100 | 20 | 5 | 11.5 |

# Precision and Recall

## Averaging

- Interpolated precision-recall curves



# Precision and Recall

## Recall-Precision Curve

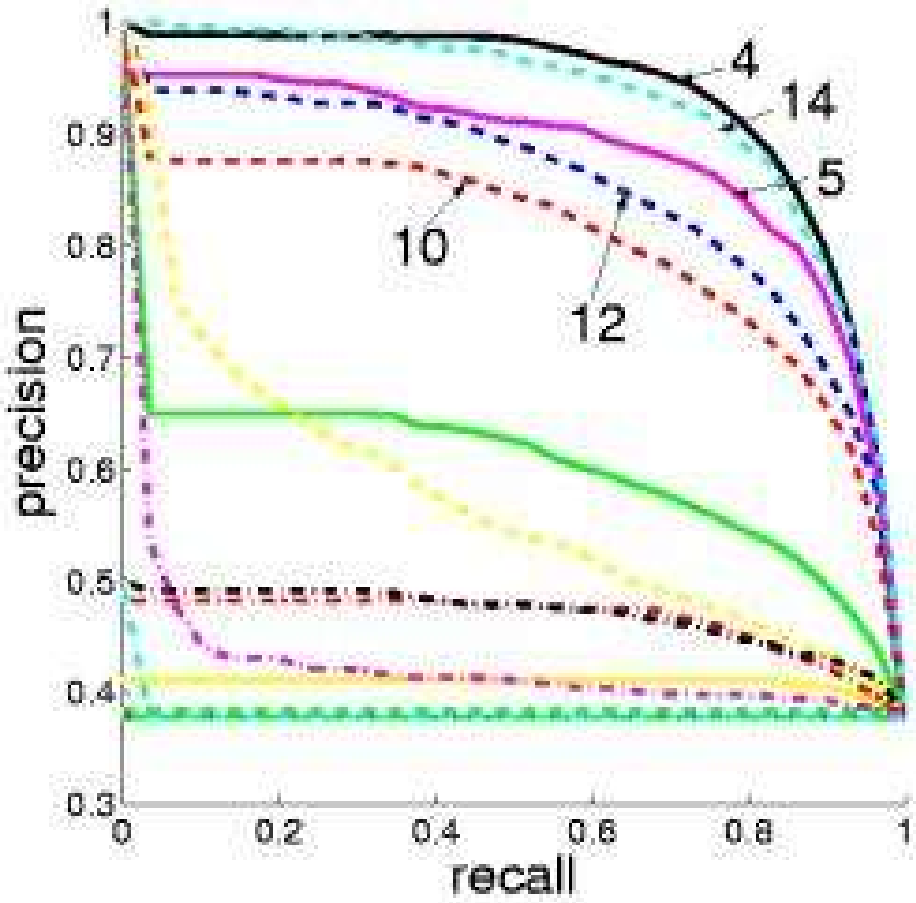
- Advantages
  - Simple, intuitive, and combined in single curve
  - Provide quantitative evaluation of the answer set and comparison among retrieval algorithms
  - A standard evaluation strategy for IR systems

- Disadvantages
  - Can’t know true recall value except in small document collections
  - Assume a strict document rank ordering

# Precision and Recall

## Comparison of systems

- We can now compare IR systems (or versions)



# Precision and Recall

## The complete methodology

To evaluate an IR system (or version):

1. For each query in the test collection
   1) Run the query against the system to obtain a ranked list of retrieved documents
   2) Use the ranking and relevance judgements to calculate recall/precision pairs
2. Average recall / precision values across all queries to obtain an overall measure of the effectiveness

# Precision and Recall

## Problem with the test collection methodology

- Building larger test collections along with complete relevance judgement is difficult or impossible
  - require assessor time - very expensive
  - Require many diverse retrieval runs.
- Recall is difficult if not impossible to get correctly as there is no way we can find all the relevant documents for each query
- Issues:
  - Non-judged documents are assumed non-relevant
  - Can we reuse the test collection later on?

# Single Value Summaries

## Precision at document cut-off level

| Document Level Average | |
|---|---|
| Recall | Precision |
| At 5 documents | 0.42 |
| At 10 documents | 0.40 |
| ... | ... |
| At 1000 documents | 0.05 |

- Actual performance as a user might see it
- often used in web retrieval, usually written P@i; e.g. P@10

# Single Value Summaries

## R-Precision

- Generate a single value summary of ranking by computing the precision at the R-th position in the ranking
  - Where R is the total number of relevant docs for the current query

1. $d_{123}$ ●
2. $d_{84}$
3. $d_{56}$ ● ■
4. $d_{6}$
5. $d_{8}$
6. $d_{9}$ ●
7. $d_{511}$
8. $d_{129}$ ■
9. $d_{187}$
10. $d_{25}$ ●
11. $d_{38}$
12. $d_{48}$
13. $d_{250}$
14. $d_{113}$
15. $d_{3}$ ● ■

$R_q=\{d_3, d_5, d_9, d_{25}, d_{39}, d_{44}, d_{56}, d_{71}, d_{89}, d_{123}\}$

- 10 relevant documents (●)

=> *R*-precision = 4/10=0.4

$R_q=\{d_3, d_{56}, d_{129}\}$

- 3 relevant document (■)

=> *R*-precision=1/3=0.33

# Single Value Summaries

## Mean Average Precision (mAP)

- Averaged at relevant docs and across queries
  - E.g. relevant docs ranked at 1, 5, 10, precisions are 1/1, 2/5, 3/10,
    - non-interpolated average precision
  - Mean Average Precision (denoted as mAP or MAP)

$$\frac{1}{|Q|}\sum_{q=1}^{|Q|}(\text{non}-\text{interpoloated average precision})_q$$

  - Widely used in IR performance evaluation

# Alternative Measures

## The Harmonic Mean (F Measure)

- The harmonic mean F of recall and precision
  - r(j): the recall for the j-th document in the ranking
  - P(j): the precision for the j-th document in the ranking

$$F(j) = \frac{2}{\frac{1}{r(j)} + \frac{1}{P(j)}} = \frac{2 \cdot P(j) \cdot r(j)}{P(j) + r(j)}$$

- Characteristics
  - F = 0: no relevant documents were retrieved
  - F = 1: all ranked documents are relevant
  - A high F achieved only when both recall and precision are high
  - Determination of the maximal F can be interpreted as an attempt to find the best possible compromise between recall and precision

# Alternative Measures

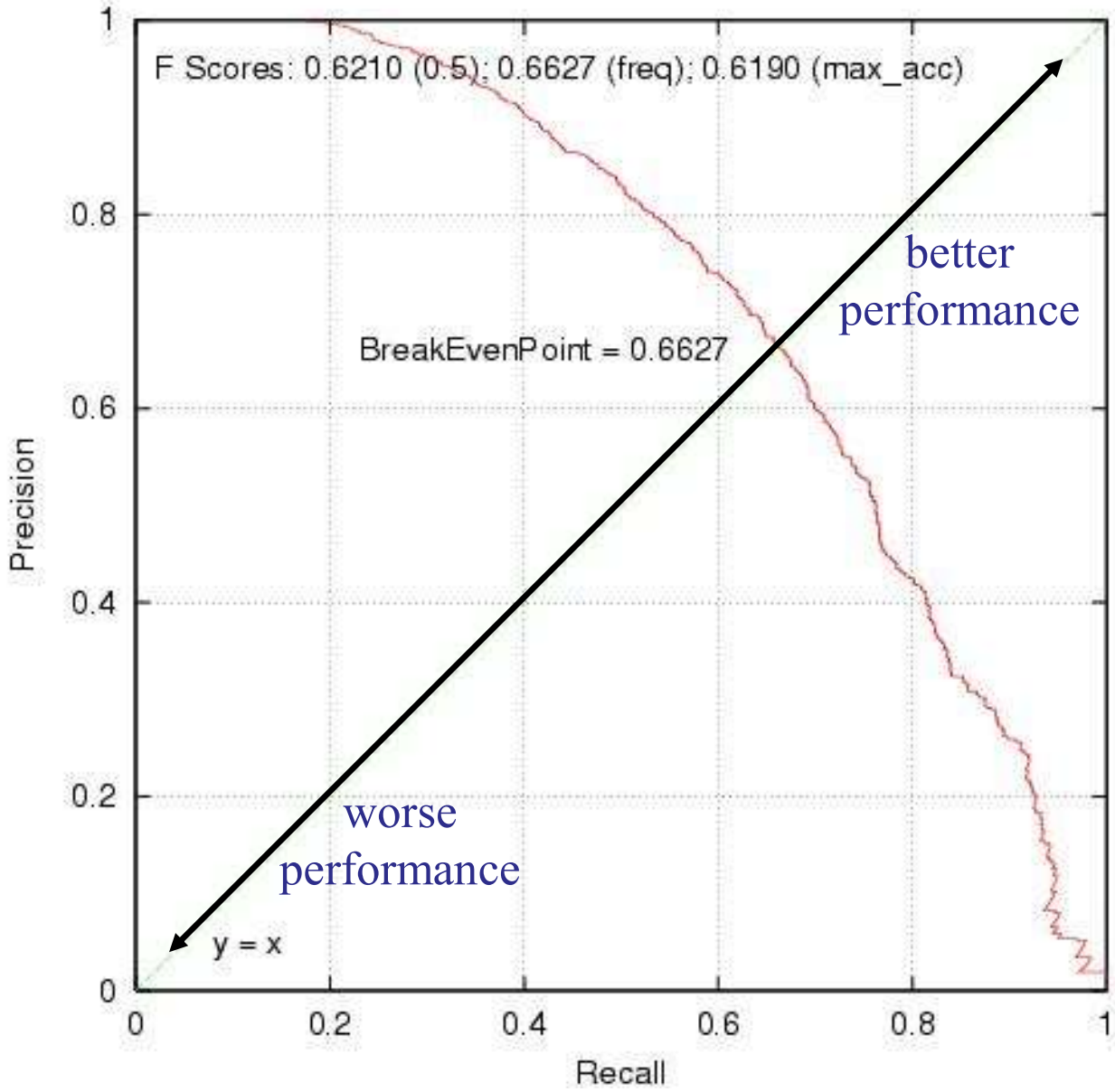
## Break-even Point

$$F = \frac{2 * (PRECISION \times RECALL)}{(PRECISION + RECALL)}$$

harmonic average of precision and recall

$$BreakEvenPoint = PRECISION = RECALL$$

# Alternative Measures




Use the best F score to represent this P-R curve:

**the break-even point**

# Alternative Measures

## The E Measure

- Another measure which combines recall and precision
- Allow the user to specify whether he/she is more interested in recall or precision

$$E(j) = 1 - \frac{1 + b^2}{\frac{b^2}{r(j)} + \frac{1}{P(j)}} = 1 - \frac{(1 + b^2) \cdot P(j) \cdot r(j)}{b^2 \cdot P(j) + r(j)}$$

- Characteristics
  - b = 1: act as the complement of F Measure
  - b > 1: more interested in recall
  - b < 1: more interested in precision

# Alternative Measures

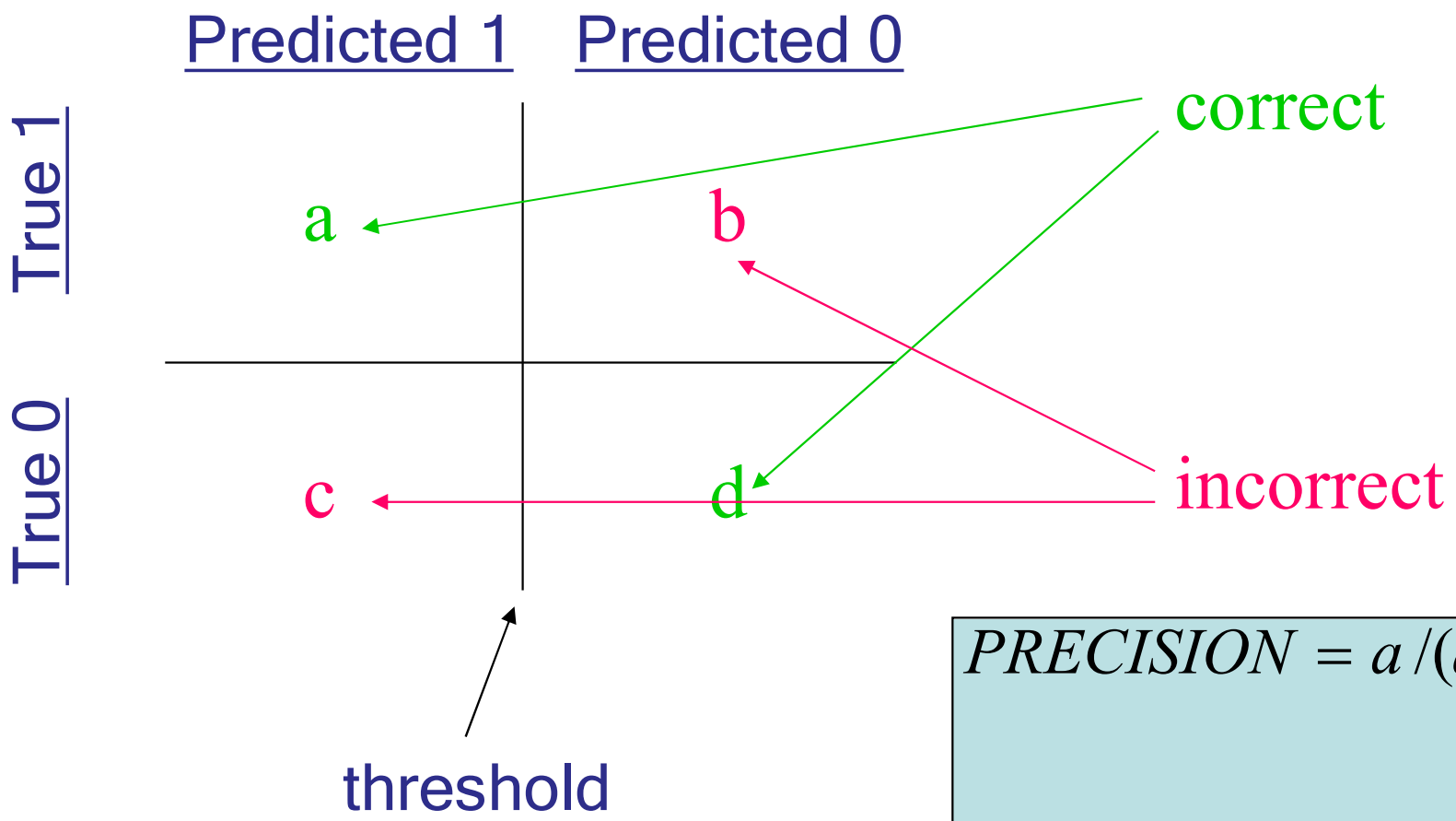
## Receiver Operating Characteristics

## ROC Plot and ROC Area

- Developed in WWII to statistically model false positive and false negative detections of radar operators
- Better statistical foundations than most other measures
- Standard measure in medicine and biology
- Becoming more popular in ML

# Alternative Measures

## Confusion matrix




a: TP
b: FN
c: FP
d: TN

$$PRECISION = a/(a+c)$$

$$RECALL = a/(a+b)$$

# Alternative Measures

## ROC Plot

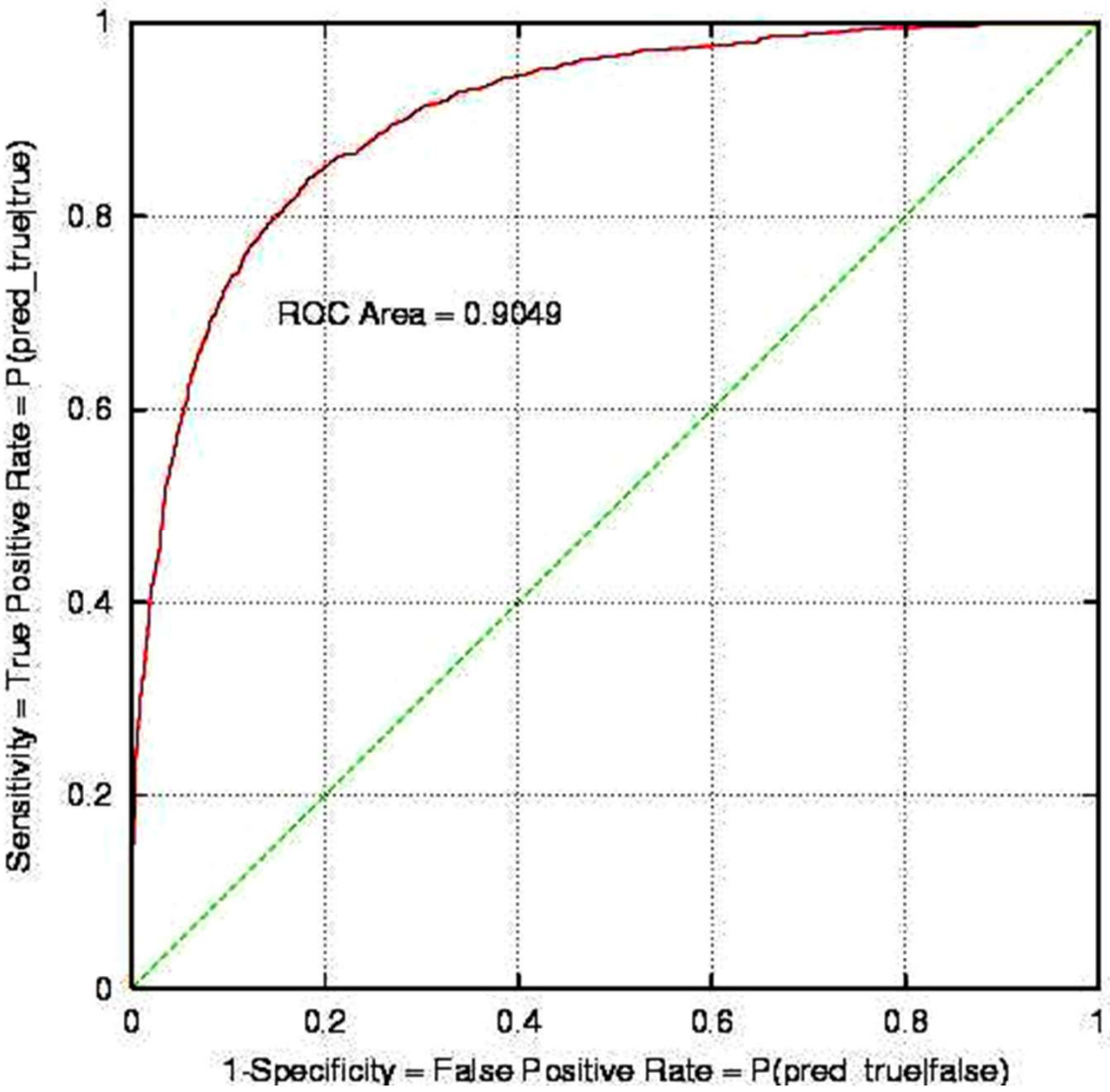
- Plot
  - TP rate vs. FP rate
  - P(true|true) vs. P(true|false)
- Sensitivity = a/(a+b) = Recall
- 1 - Specificity = 1 - d/(c+d)=1-TN/(FP+TN)
- Sensitivity vs. 1-Specificity

# Alternative Measures

## Properties of ROC

ROC Area:

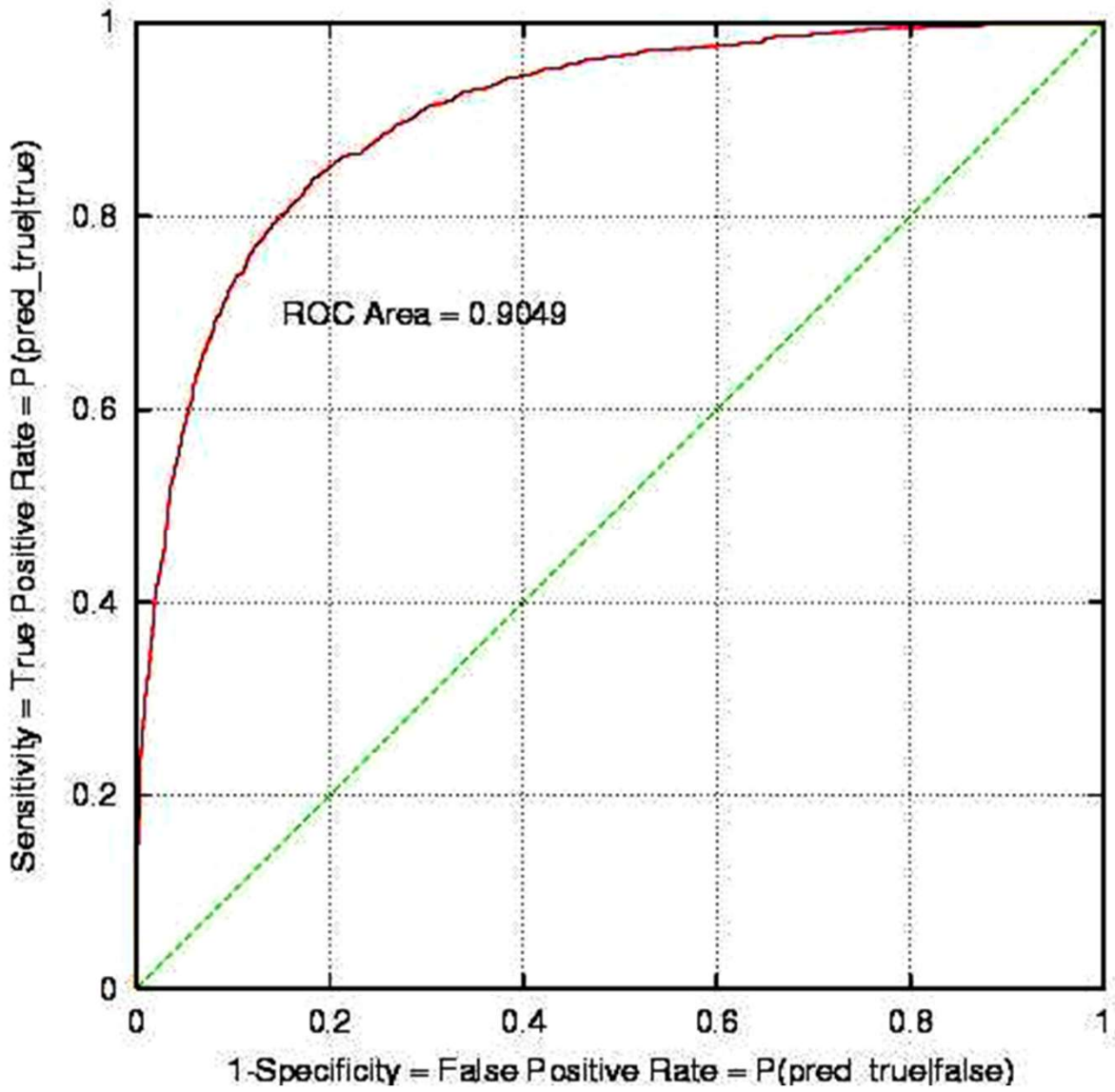
- 1.0: perfect prediction
- 0.9: excellent prediction
- 0.8: good prediction
- 0.7: mediocre prediction
- 0.6: poor prediction
- 0.5: random prediction
- <0.5: something wrong!



# Alternative Measures

## Properties of ROC

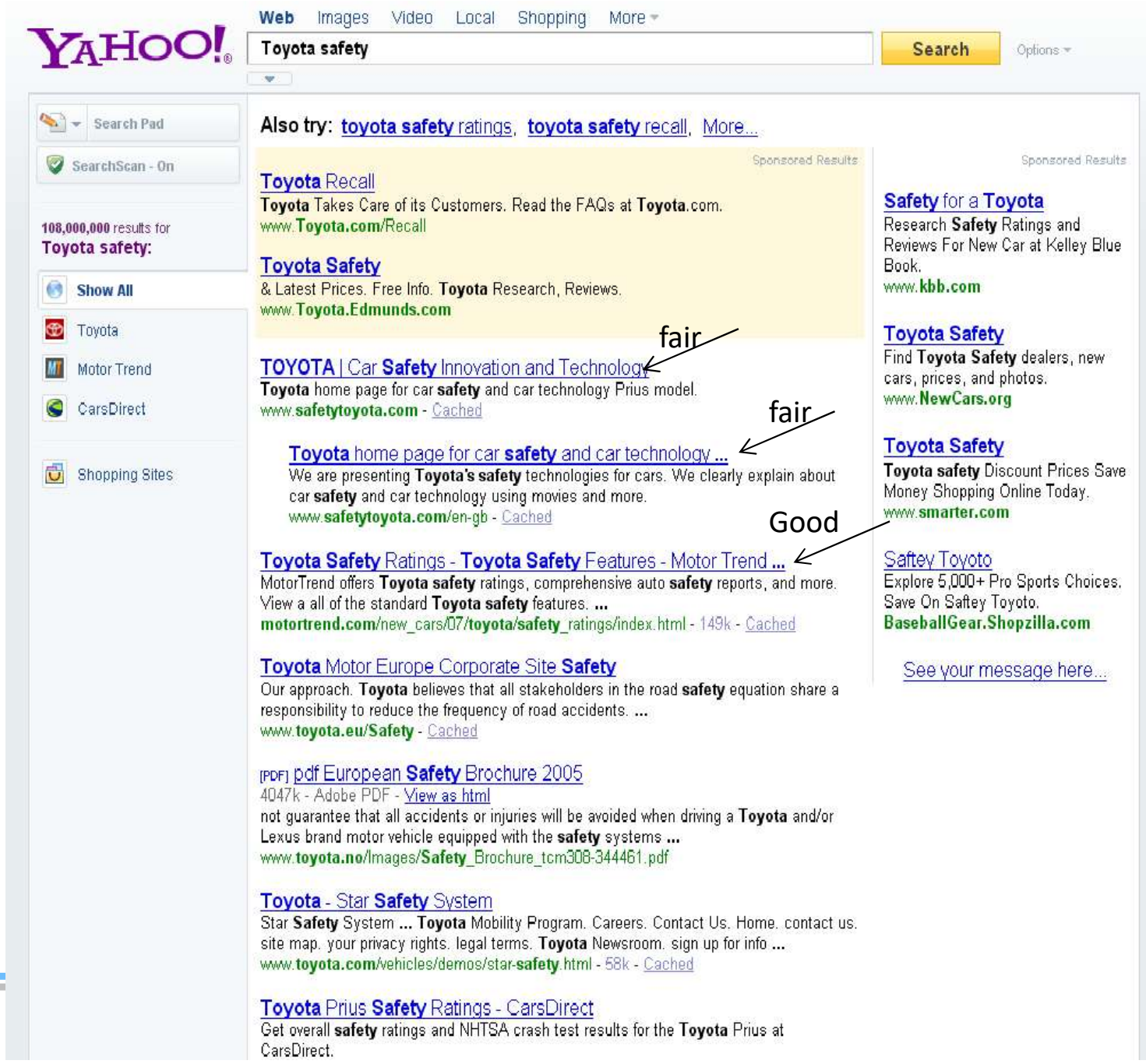
- Slope is non-increasing
- Each point on ROC represents different tradeoff (cost ratio) between false positives and false negatives
- Slope of line tangent to curve defines the cost ratio
- ROC Area represents performance averaged over all possible cost ratios



# Beyond Binary Relevance

- Precision and recall allow only binary relevance assessments
- As a result, there is no distinction between highly relevant docs and mildly relevant docs
- These limitations can be overcome by adopting graded relevance assessments and metrics that combine them

# Beyond Binary Relevance



# Discounted Cumulative Gain

- The discounted cumulated gain (DCG) is a metric that combines graded relevance assessments effectively
- Popular measure for evaluating web search and related tasks

- Two assumptions:
  - Highly relevant documents are more useful than marginally relevant documents
  - the lower the ranked position of a relevant document, the less useful it is for the user, since it is less likely to be examined

# Discounted Cumulative Gain

## Summarize a Ranking: DCG

- What if relevance judgments are in a scale of [0,$r$ ]?
- Cumulative Gain (CG) at rank $p$
  - Let the ratings of the $n$ documents be $r_1$, $r_2$, …$r_p$ (in ranked order)
  - $CG_p = r_1+r_2+\ldots r_p$
- Discounted Cumulative Gain (DCG) at rank $p$
  - $DCG_p = r_1 + r_2/\log_2 2 + r_3/\log_2 3 + \ldots r_p/\log_2 p$
    - We may use any base for the logarithm

# Discounted Cumulative Gain

- *DCG* is the total gain accumulated at a particular rank *p*:

$$DCG_p = rel_1 + \sum_{i=2}^{p} \frac{rel_i}{\log_2 i}$$

- Alternative formulation:

$$DCG_p = \sum_{i=1}^{p} \frac{2^{rel_i} - 1}{log(1+i)}$$

  - used by some web search companies
  - emphasis on retrieving highly relevant documents

# Discounted Cumulative Gain

## DCG Example

- 10 ranked documents judged on 0-3 relevance scale:
  3, 2, 3, 0, 0, 1, 2, 2, 3, 0
- DG= $r_i/\log_2 i$ $(i>=2)$

  = 3, 2, 1.89, 0, 0, 0.39, 0.71, 0.67, 0.95, 0

- DCG= $r_1 + r_2/\log_2 2 + r_3/\log_2 3 + \ldots r_p/\log_2 p$

  = 3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61

# Discounted Cumulative Gain

## Summarize a Ranking: NDCG

- Normalized Discounted Cumulative Gain (NDCG) at rank *p*
  - Normalize DCG at rank *p* by the DCG value at rank *p* of the ideal ranking
  - The ideal ranking would first return the documents with the highest relevance level, then the next highest relevance level, etc
- Normalization useful for contrasting queries with varying numbers of relevant results
- NDCG is now quite popular in evaluating Web search

# Discounted Cumulative Gain

## Summarize a Ranking: NDCG

- Perfect ranking:
  - 3, 3, 3, 2, 2, 2, 1, 0, 0, 0
- ideal DCG values:
  - 3, 6, 7.89, 8.89, 9.75, 10.52, 10.88, 10.88, 10.88, 10.88
- Actual DCG for the actual ranking (3, 2, 3, 0, 0, 1, 2, 2, 3, 0):
  - 3, 5, 6.89, 6.89, 6.89, 7.28, 7.99, 8.66, 9.61, 9.61
- NDCG values (divide actual by ideal):

1, 0.83, 0.87, 0.76, 0.71, 0.69, 0.73, 0.8, 0.88, 0.88

- NDCG $\leq$ 1 at any rank position

# NDCG - Example

4 documents: $d_1$, $d_2$, $d_3$, $d_4$

| i | Ground Truth | | Ranking Function$_1$ | | Ranking Function$_2$ | |
|---|---|---|---|---|---|---|
| | Doc. Order | $r_i$ | Doc. Order | $r_i$ | Doc. Order | $r_i$ |
| 1 | d4 | 2 | d3 | 2 | d3 | 2 |
| 2 | d3 | 2 | d4 | 2 | d2 | 1 |
| 3 | d2 | 1 | d2 | 1 | d4 | 2 |
| 4 | d1 | 0 | d1 | 0 | d1 | 0 |
| | $NDCG_{GT}$=1.00 | | $NDCG_{RF1}$=1.00 | | $NDCG_{RF2}$=0.9203 | |

$$DCG_{GT} = 2 + \left( \frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.6309$$

$$DCG_{RF2} = 2 + \left( \frac{1}{\log_2 2} + \frac{2}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.2619$$

$$DCG_{RF1} = 2 + \left( \frac{2}{\log_2 2} + \frac{1}{\log_2 3} + \frac{0}{\log_2 4} \right) = 4.6309$$

$$MaxDCG = DCG_{GT} = 4.6309$$

# Discounted Cumulative Gain

## Discussion on DCG Metrics

- CG and DCG metrics aim at taking into account multiple level relevance assessments
- This has the advantage of distinguishing highly relevant documents from mildly relevant ones
- The inherent disadvantages are that multiple level relevance assessments are harder and more time consuming to generate

# Discounted Cumulative Gain

## Discussion on DCG Metrics

- Despite these inherent difficulties, the CG and DCG metrics present benefits:
  - They allow systematically combining document ranks and relevance scores
  - Cumulated gain (CG) provides a single metric of retrieval performance at any position in the ranking
  - It also stresses the gain produced by relevant docs up to a position in the ranking, which makes the metrics more immune to outliers
  - Further, discounted cumulated gain (DCG) allows down weighting the impact of relevant documents found late in the ranking